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FETAL CIRCULATION AND INTRODUCTION TO CONGENITAL HEART DISEASES

Fetal Circulation

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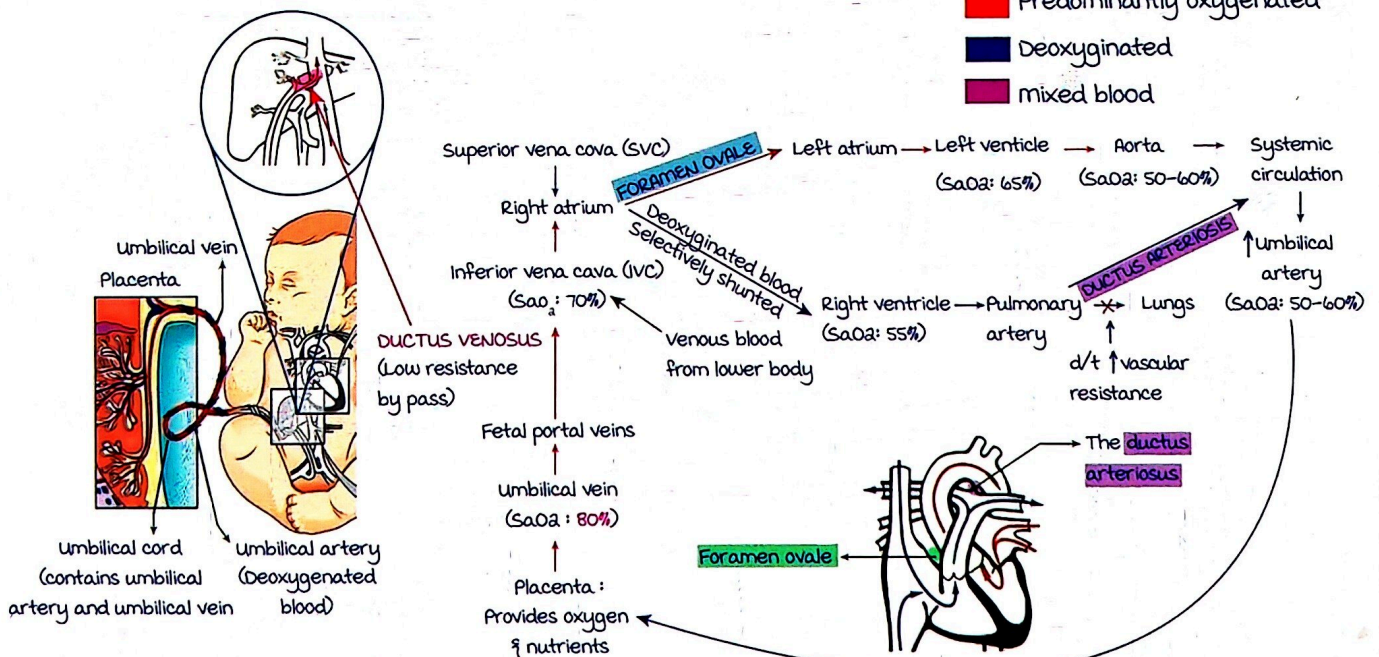
Salient features :

- Pulmonary vascular resistance : High (d/t non-functional lungs) $\uparrow$ BP in right side of heart.
- Communications in a fetus :
 - Ductus arteriosus
 - Ductus venosus
 - Foramen ovale
 - Umbilical vessels : 2 umbilical arteries + 1 umbilical vein.

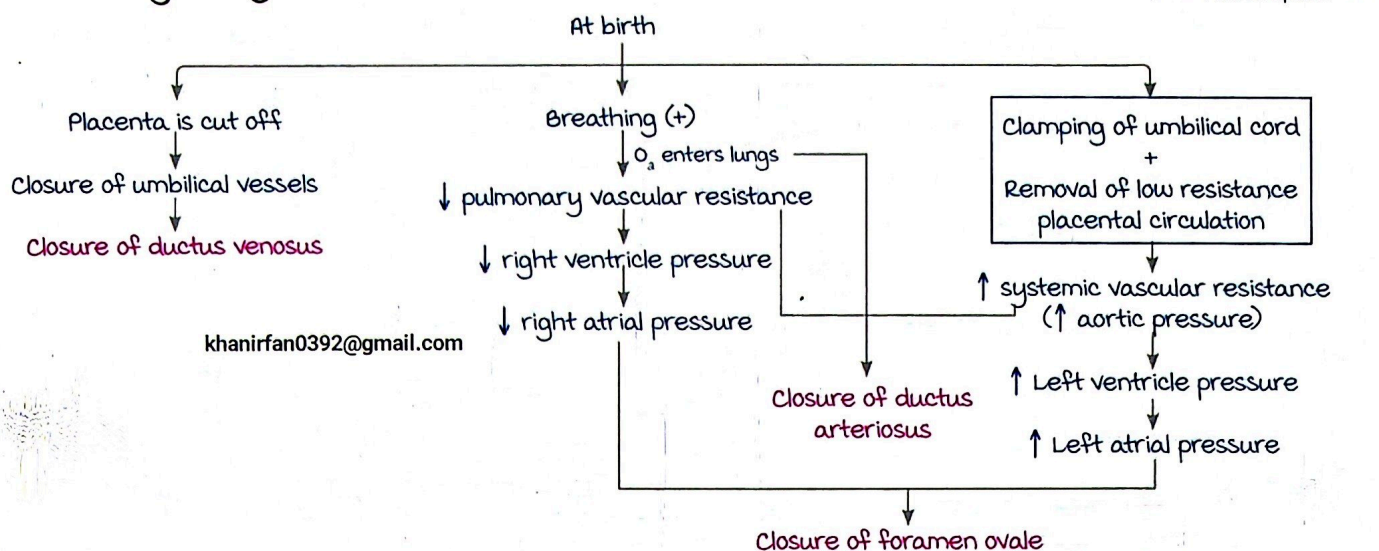
Circulation before birth :

Index :

- Predominantly oxygenated
- Deoxygenated
- mixed blood



Circulatory changes at birth :



Outcome of fetal structures :

Structure	Closure		Remnant
	Functional	Anatomical	
Ductus venosus	Soon after birth	7 days	Ligamentum venosum
Foramen ovale	Soon after birth	3 months (up to 1 year)	Fossa ovalis
Ductus arteriosus	10-15h after birth	10-21 days after birth	Ligamentum arteriosum

Congenital heart diseases (CHD)

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Prevalence : 6 to 8 / 1000 live births.

Inheritance : multifactorial.

most common CHD : Ventricular septal defect (VSD).

most common cyanotic CHD : Tetralogy of Fallot (TOF)

ASSOCIATIONS :

Genetic		maternal	
Syndrome	Defects	Condition	Defects
Down Syndrome	• Endocardial cushion (Atrio-ventricular septal defect).	Overt's diabetes	• VSD (m/c). • Transposition of great arteries (most specific).
Turner Syndrome	• Bicuspid aortic valve. • Coarctation of aorta.	Rubella	• Patent ductus arteriosus. • PS. • VSD.
Noonan Syndrome	• Pulmonary stenosis. • Atrial septal defects (ASD).	Systemic lupus erythematosus (SLE)	Heart block d/t: • Anti-Ro (SS-A). • Anti-La (SS-B).

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Genetic		maternal	
Syndrome	Defects	Condition	Defects
DiGeorge Syndrome (CATCH22 : microdeletion)	• TOF. • Conotruncal defects.	Teratogens : a. Lithium. b. Phenytoin. c. Alcohol.	a. Ebstein's anomaly. b. VSD. c. VSD.
Williams Syndrome : (microdeletion)	• Supravalvular aortic stenosis. • Elfin facies.		
Holt-Oram syndrome : (Autosomal dominant)	• ASD. • Skeletal anomalies (Eg : Absent radius thumb).		
Alagille Syndrome	• Pulmonary stenosis (PS). • Triangular face & cholestasis.		

NADA'S CRITERIA FOR CHD

Diagnosis : Presence of 1 major OR 2 minor criteria.

major criteria	minor criteria:
1. Systolic murmur : $\geq$ Grade 3. 2. Any diastolic murmur. 3. Central cyanosis with no improvement on O ₂ inhalation (Improvement $\rightarrow$ Respiratory etiology). 4. Features of congestive cardiac failure (CCF).	1. Systolic murmur ($<$ Grade 3). 2. Abnormal BP. 3. Abnormal ECG. 4. Abnormal X-ray. 5. Abnormal Sa.